



IdentityWallet

USER MANUAL

How to use everything

Step-by-step guide to installing, running and using every feature of the Digital Identity Wallet, for holders, issuers, administrators and verifiers.

Installation

Wallet holder

Issuer

Admin

Verifier

Troubleshooting

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Version	1.0

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1 Getting Started

1.1 What is IdentityWallet?

IdentityWallet keeps your identity documents as **digitally signed credentials** — like a national ID card, driving licence or student ID, but issued electronically by the organisation and impossible to forge. When someone needs to check something about you, you share **only what they need** (for example "Age over 18: Yes") using a link or QR code, and you can see every time they open it.

1.2 Who uses it

Role	Who they are	What they do
Wallet holder	Any person (student, citizen, employee)	Keeps credentials, shares selected details, stores documents in the vault
Issuer staff	Staff of a trusted organisation (govt. office, college, RTO)	Issues credentials into holders' wallets and revokes them
Administrator	Platform operator	Registers issuers, manages users, checks the audit log
Verifier	Bank, cinema, employer...	Opens a link or scans a QR to check details — no account needed

1.3 Demo accounts

The first time the backend starts, it creates these accounts:

Role	Email	Password
Wallet holder	<code>aarav@wallet.demo</code>	<code>User@1234</code>
Wallet holder (empty wallet)	<code>priya@wallet.demo</code>	<code>User@1234</code>
Issuer — Government	<code>registrar@identity.demo</code>	<code>Issuer@123</code>
Issuer — College	<code>admissions@dit.demo</code>	<code>Issuer@123</code>
Issuer — Transport office	<code>licensing@rto.demo</code>	<code>Issuer@123</code>
Administrator	<code>admin@idwallet.local</code>	<code>Admin@123</code>

Important

These passwords are for demonstration only. Change them in `backend/src/main/resources/application.properties` before any real use.

2 Installation and Running

2.1 Requirements

Software	Version	How to check
Java (JDK)	17 or newer	<code>java -version</code>
Node.js + npm	18 or newer	<code>node -v</code>
Browser	Chrome, Edge or Firefox	—
Maven	Not needed — <code>mvnw.cmd</code> downloads it automatically	—

If Java is missing, install **JDK 21** from adoptium.net and tick "**Set JAVA_HOME variable**" during setup. Then open a **new** PowerShell window.

2.2 First-time setup

- 1 Open PowerShell and go to the backend folder:

```
cd "D:\Web-Development\Java Projects\identity-wallet\backend"
```

- 2 (Optional) run the automated tests. The first run downloads Maven and libraries, so it takes a few minutes:

```
.\mvnw.cmd test
```

You should see `Tests run: 8, Failures: 0` and `BUILD SUCCESS`.

- 3 Open a second PowerShell window for the website and install its packages (only once):

```
cd "D:\Web-Development\Java Projects\identity-wallet\frontend"
npm install
```

2.3 Starting the project (every time)

- 1 **Window 1 — backend:**

```
cd "D:\Web-Development\Java Projects\identity-wallet\backend"
.\mvnw.cmd spring-boot:run
```

Wait for `Started IdWalletApplication`. Keep this window open.

- 2 **Window 2 — website:**

```
cd "D:\Web-Development\Java Projects\identity-wallet\frontend"
npm run dev
```

3 Open **http://localhost:5173** in your browser.

To stop, press **Ctrl + C** in each window.

2.4 Useful addresses

Address	What it is
<code>http://localhost:5173</code>	The website (use this)
<code>http://localhost:8080/h2-console</code>	Database viewer — JDBC URL <code>jdbc:h2:file:./data/idwallet</code> , user <code>sa</code> , empty password
<code>http://localhost:5173/verify</code>	Public verifier page

2.5 Resetting the demo

Stop the backend and delete the folder `backend\data`. On the next start a fresh database, a new master key and the demo data are created.

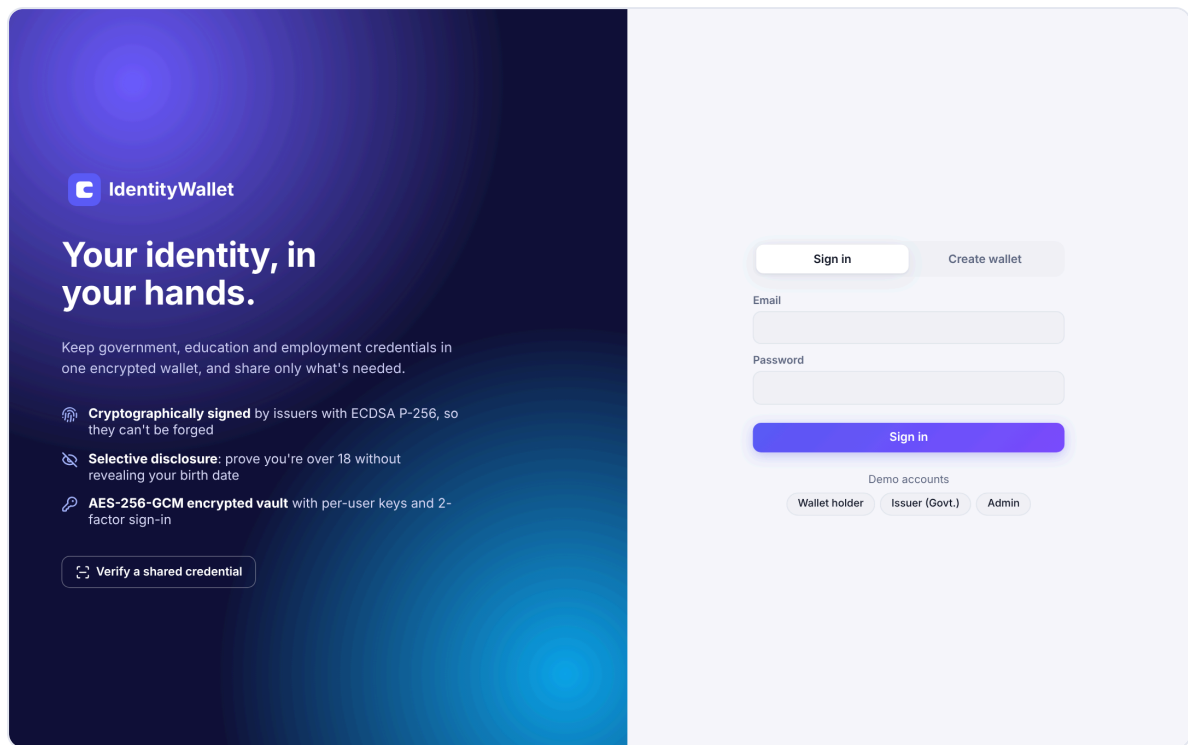
Never delete only master.key

`backend\data\master.key` decrypts all vault documents and keys. Delete the whole `data` folder or nothing. Never upload it to GitHub (the `.gitignore` already excludes it).

2.6 Using MySQL instead of H2 (optional)

1. Create a database: `CREATE DATABASE idwallet;`
2. Set your MySQL username/password in `application-mysql.properties`.
3. Start with: `.\mvnw.cmd spring-boot:run "-Dspring-boot.run.profiles=mysql"`

3 Signing In and Creating a Wallet



IdentityWallet

Your identity, in your hands.

Keep government, education and employment credentials in one encrypted wallet, and share only what's needed.

- Cryptographically signed** by issuers with ECDSA P-256, so they can't be forged
- Selective disclosure:** prove you're over 18 without revealing your birth date
- AES-256-GCM encrypted vault** with per-user keys and 2-factor sign-in

[Verify a shared credential](#)

[Sign in](#) [Create wallet](#)

Email

Password

[Sign in](#)

Demo accounts

[Wallet holder](#) [Issuer \(Govt.\)](#) [Admin](#)

Sign-in page

3.1 Sign in

1. Enter your email and password, or click a **Demo accounts** chip (it fills both fields).
2. Click **Sign in**. You land on the page for your role.
3. If two-factor sign-in is on, enter the **6-digit code** from your authenticator app and click **Verify**.

3.2 Create a new wallet

IdentityWallet

Your identity, in your hands.

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- Cryptographically signed** by issuers with ECDSA P-256, so they can't be forged
- Selective disclosure:** prove you're over 18 without revealing your birth date
- AES-256-GCM encrypted vault** with per-user keys and 2-factor sign-in

[Verify a shared credential](#)

[Sign in](#) [Create wallet](#)

Full name

Email

Password

At least 8 characters with a letter, a digit and a symbol. Your wallet gets its own signing key and DID.

[Create my wallet](#)

Create wallet tab

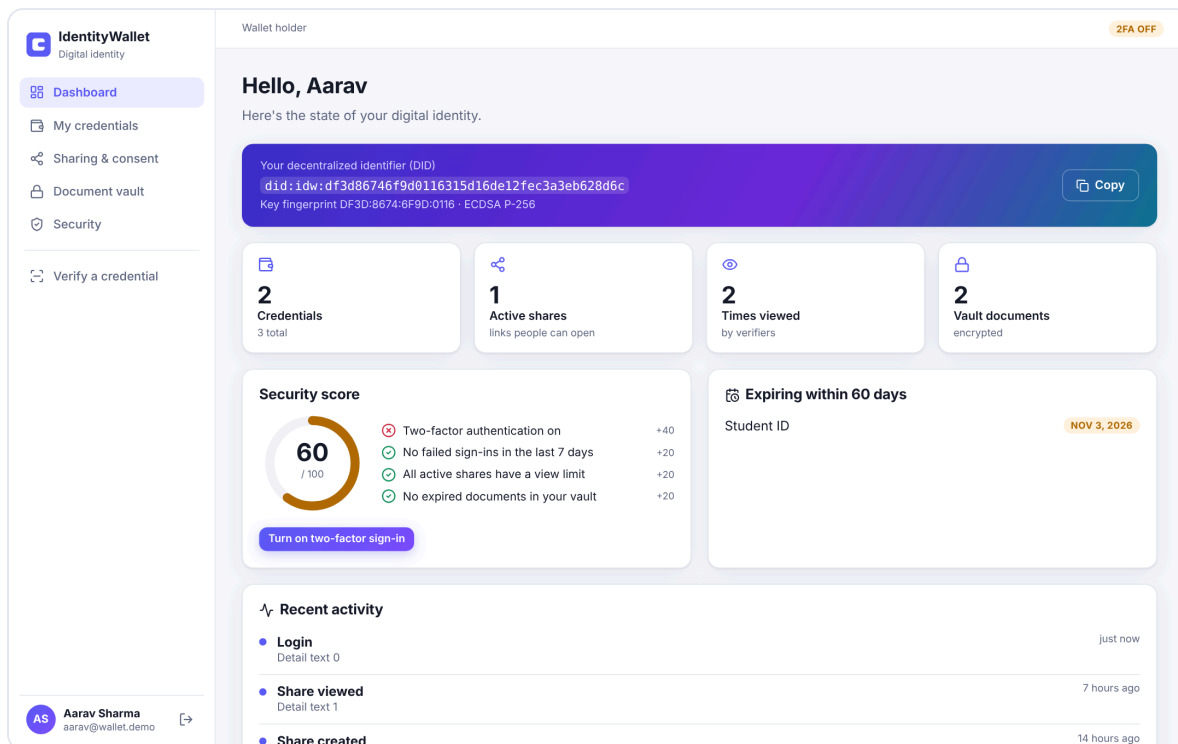
1. Click the **Create wallet** tab.
2. Enter your full name, email and a password with **at least 8 characters, a letter, a digit and a symbol**.
3. Click **Create my wallet**. Your account receives its own signing key and a **DID** (decentralized identifier), and you are signed in.

Security behaviour you may notice

- After **5 wrong passwords** the account is locked for **5 minutes**.
- The message is always "Wrong email or password" so nobody can find out which emails are registered.
- Your session ends when you close the browser tab.

4 Wallet Holder Guide

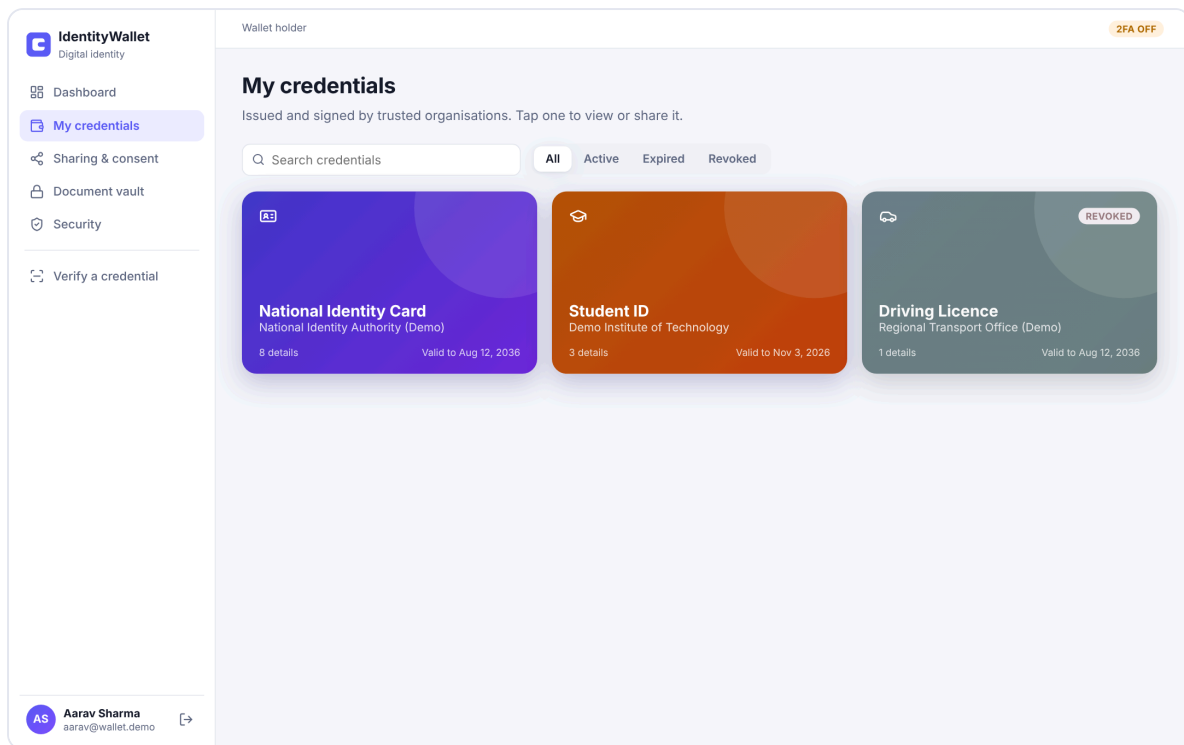
4.1 Dashboard



Holder dashboard

Area	Meaning
Purple DID card	Your decentralized identifier and key fingerprint. Copy puts the DID on the clipboard
Credentials	Active credentials (total shown below). Click to open the wallet
Active shares / Times viewed	Links currently open, and how many times verifiers opened them
Vault documents	Encrypted documents stored
Security score	Out of 100, with a checklist. Turning on 2FA gives +40
Expiring within 60 days	Credentials or vault documents about to expire
Recent activity	Your latest events (logins, shares, views). Failed attempts show a red dot

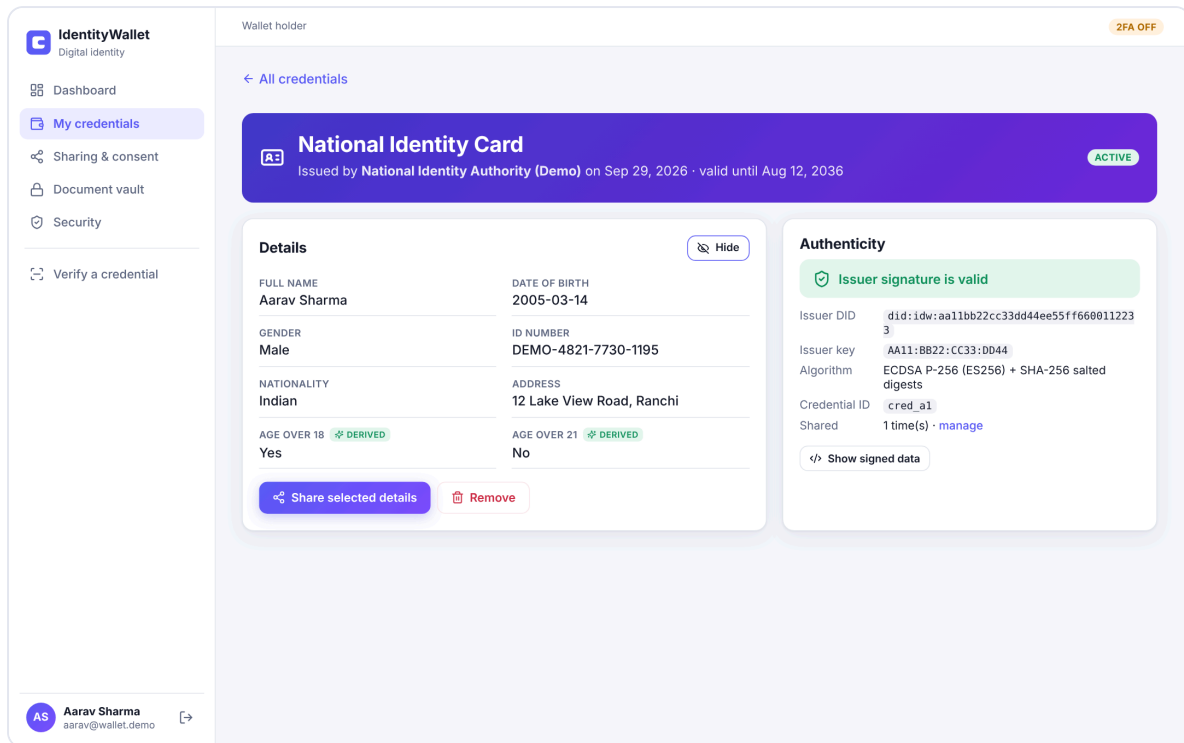
4.2 My credentials



Wallet

- Each card is one credential with its issuer, number of details and validity date.
- Use the **search box** or the **All / Active / Expired / Revoked** filter.
- A grey card with **REVOKED** has been withdrawn by its issuer and cannot be shared.

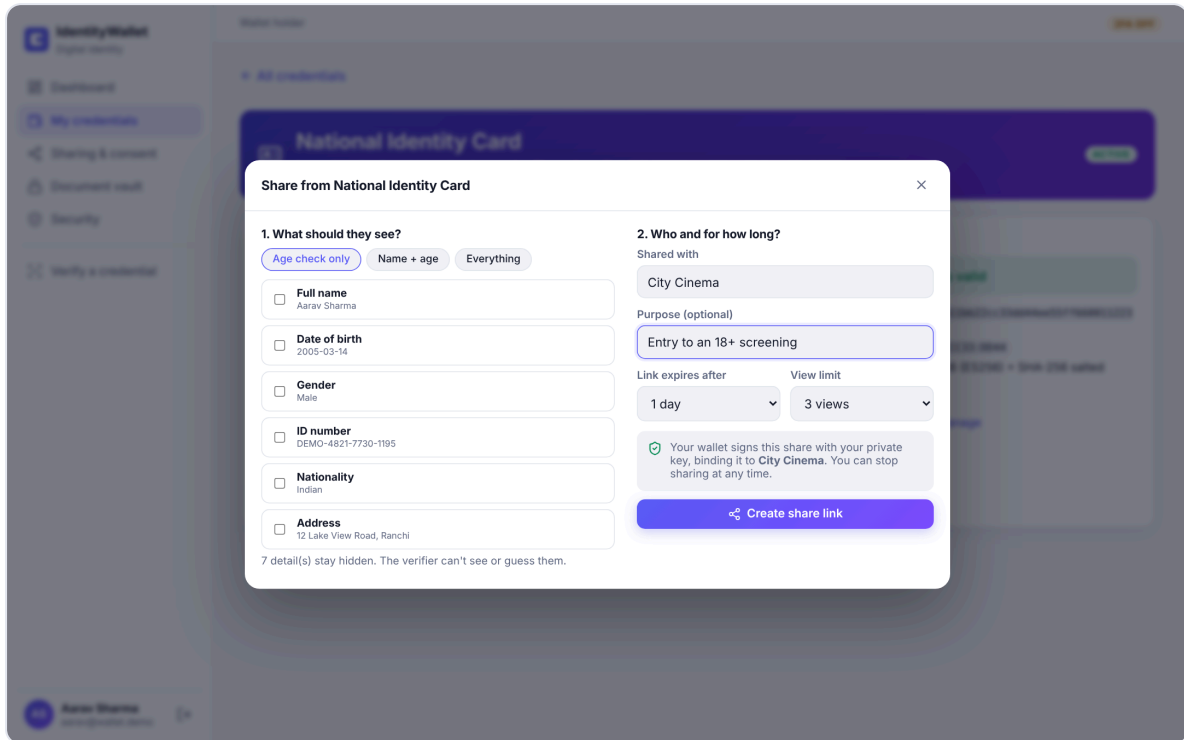
4.3 Viewing a credential



Credential details

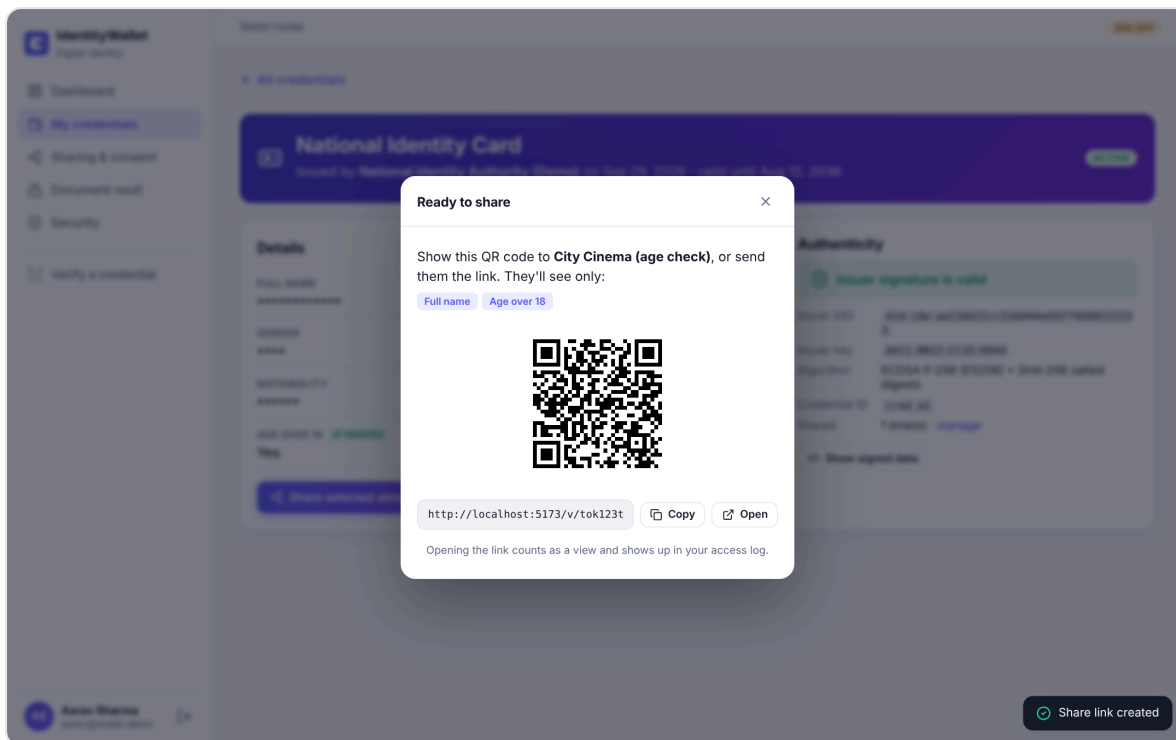
1. Click a card to open it. Values are hidden as dots for privacy.
2. Click **Reveal** to show them, **Hide** to mask again.
3. Details marked **DERIVED** (Age over 18 / 21) were calculated by the issuer from your date of birth — you can share them without revealing the date.
4. The **Authenticity** panel shows whether the issuer's signature is valid, the issuer's DID and key, and how many times this credential has been shared.
5. **Show signed data** displays exactly what the issuer signed. Notice it contains only hashes (`_sd`), never your actual details.
6. **Remove** deletes the credential from your wallet and stops all its share links.

4.4 Sharing selected details



Share dialog

- 1 Open a credential and click **Share selected details**.
- 2 **Choose what they see**. Tick individual details or use a preset: *Age check only*, *Name + age*, *Everything*. The line at the bottom tells you how many details stay hidden.
- 3 **Shared with** — type who is receiving it (required), e.g. "HDFC Bank KYC desk".
- 4 **Purpose** — optional reason, e.g. "Opening a savings account".
- 5 **Link expires after** — 15 minutes, 1 hour, 1 day, 7 days or 30 days.
- 6 **View limit** — 1, 3, 5, 10 views or unlimited. A limit improves your security score.
- 7 Click **Create share link**.



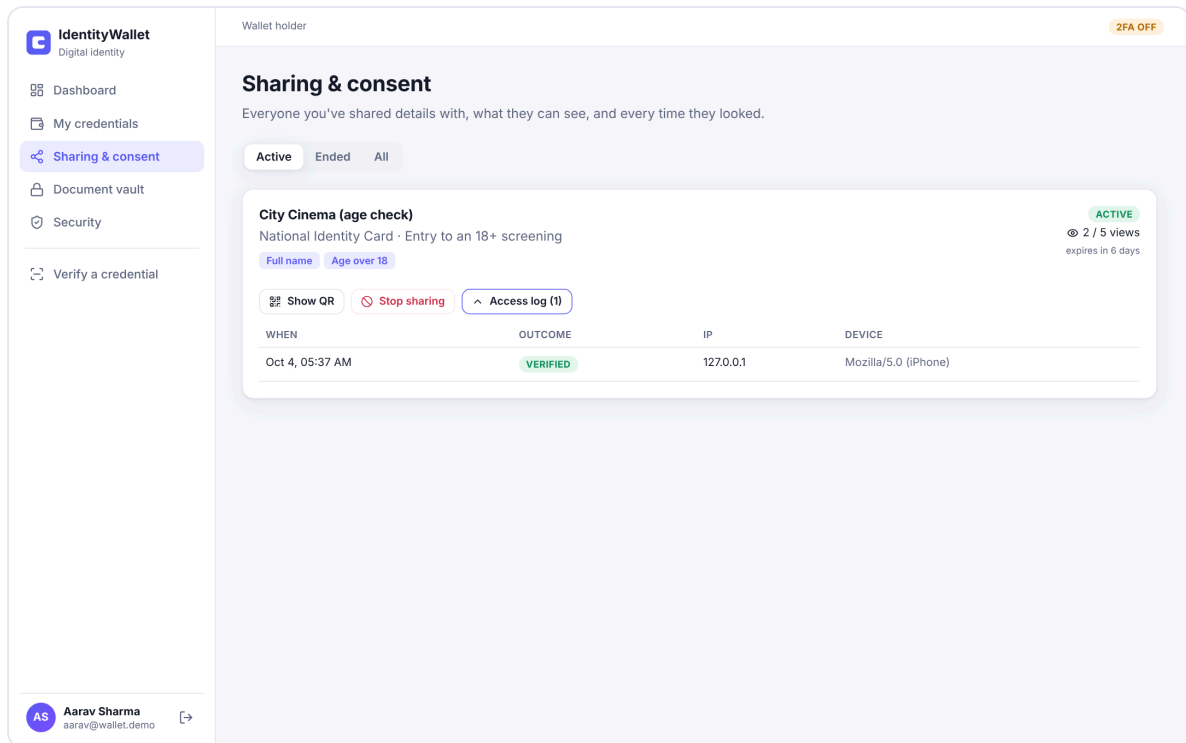
Ready to share

Now either show the **QR code** to the person, or click **Copy** and send the link. **Open** shows you what they will see (this counts as one view).

Best practice

Share the minimum. For an age check, use "Age check only". For a bank, a short expiry and 1–3 views is enough.

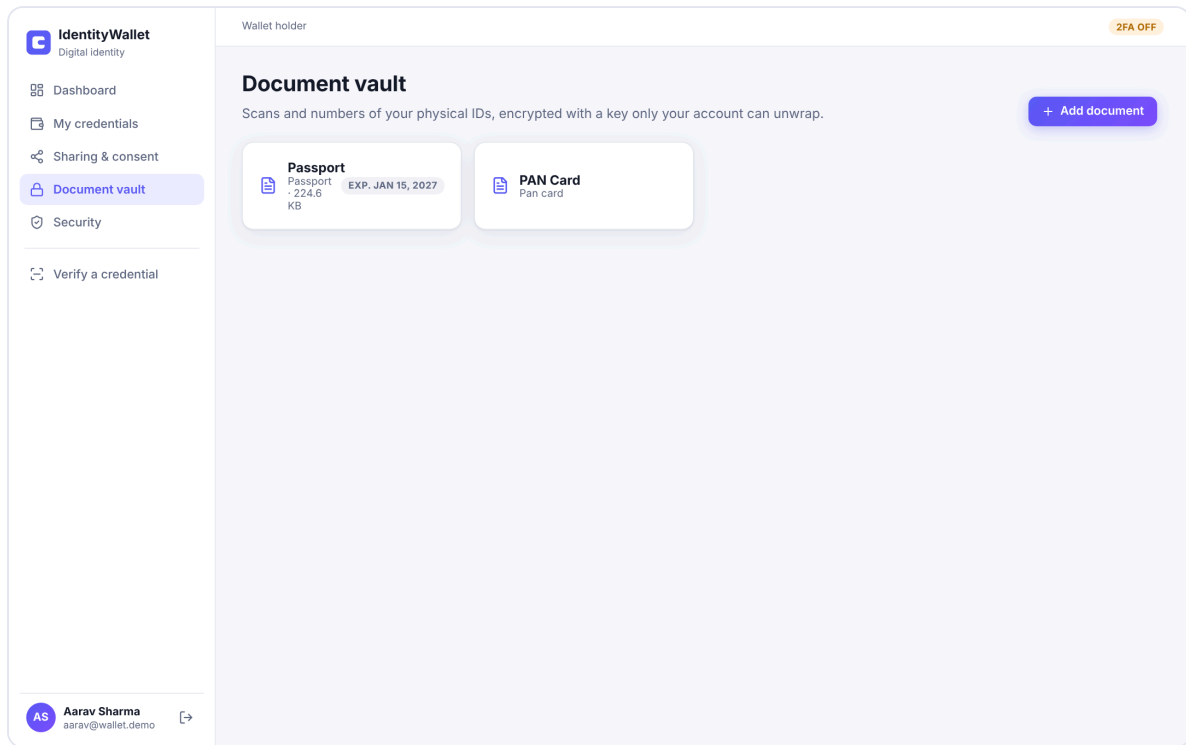
4.5 Sharing & consent page



Sharing & consent

- Tabs **Active** / **Ended** / **All** filter your shares.
- Each share shows the recipient, credential, details shared, status, views used and expiry.
- **Show QR** displays the code again. **Stop sharing** ends the link immediately.
- **Access log** lists every time the link was opened: date and time, outcome (Verified, Expired, Revoked, Used up), IP address and device.

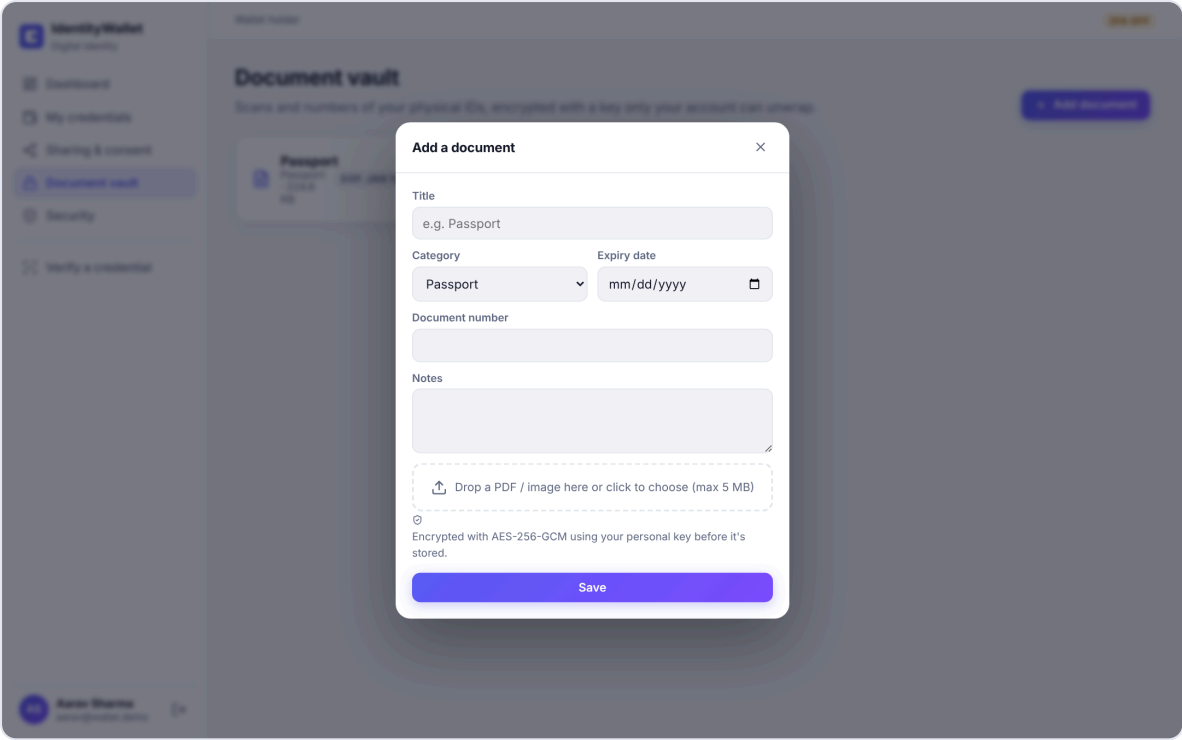
4.6 Document vault



Document vault

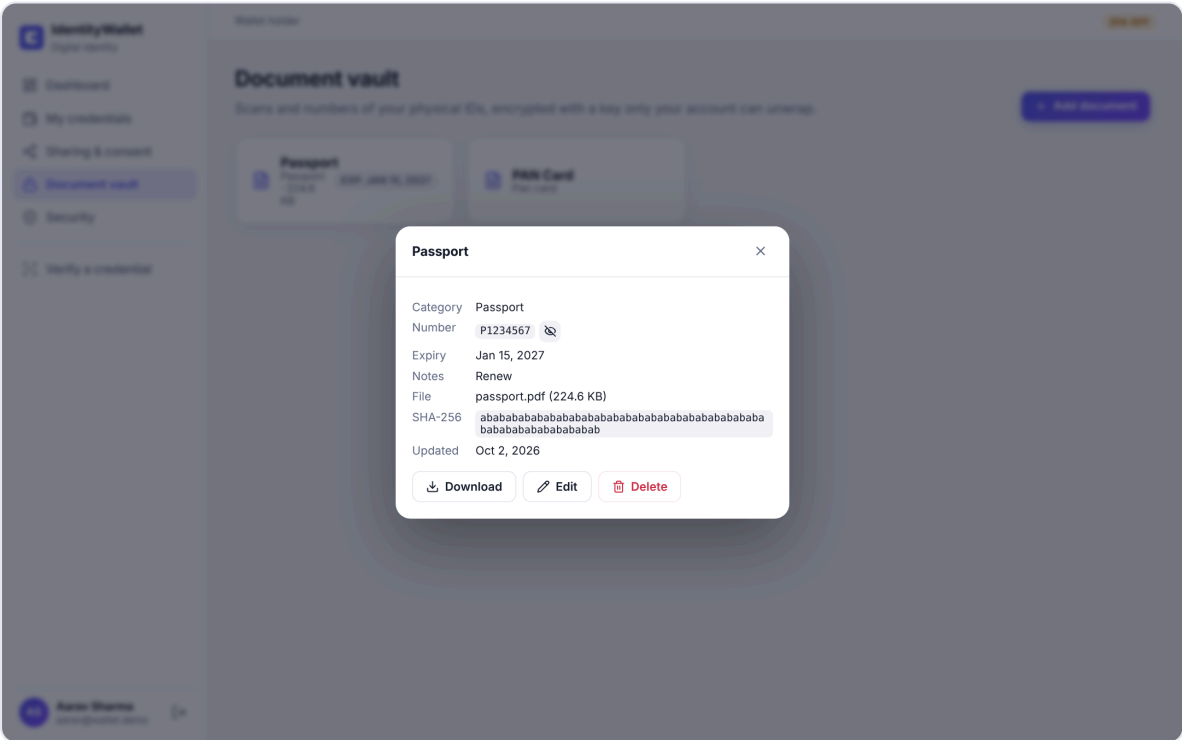
Add a document

1. Click **Add document**.
2. Fill **Title**, **Category**, optional **Expiry date**, **Document number** and **Notes**.
3. Drag a file onto the box or click it to choose one — **PDF, PNG, JPEG or WEBP**, max **5 MB**.
4. Click **Save**. Everything is encrypted with AES-256-GCM before it is stored.



Add a document

View, download, edit, delete: click a document. Use the eye icon to show the document number, **Download** to get the original file (integrity is checked with SHA-256), **Edit** to change it, **Delete** to remove it permanently. Expired documents show a red **Expired** label and lower your security score.



Document details

Why was my file rejected?

The wallet checks the real file type from its content, not from the name. A file renamed to `.pdf` that is not really a PDF is refused.

4.7 Security page

IdentityWallet

Digital identity

Dashboard

My credentials

Sharing & consent

Document vault

Security

Verify a credential

Wallet holder

2FA OFF

Security

Sign-in protection, your keys and the tamper-evident activity log.

Two-factor sign-in

Off. Anyone with your password can sign in.

Uses time-based one-time codes (TOTP, RFC 6238) that change every 30 seconds.

Set up authenticator app

Identity & keys

DID

did:dw:df3d86746f9d0116315d16de12fec3a3eb628d6c

Key fingerprint

DF3D:8674:6F9D:0116

Key type

ECDSA P-256, private key encrypted at rest

Last sign-in

Oct 4, 05:37 PM

Change password

Current password

New passwordRepeat

Update password

Activity log

Each entry stores the SHA-256 hash of the one before it, so edits to the log are detectable.

WHEN	EVENT	DETAIL	IP	HASH
Oct 4, 05:37 PM	Login	Detail text 0	127.0.0.1	ab12cd34ef...
Oct 4, 10:25 AM	Share viewed	Detail text 1	127.0.0.1	ab12cd34ef...
Oct 4, 03:13 AM	Share created	Detail text 2	127.0.0.1	ab12cd34ef...
Oct 3, 08:01 PM	Login failed	Detail text 3	127.0.0.1	ab12cd34ef...
Oct 3, 12:49 PM	Credential received	Detail text 4	127.0.0.1	ab12cd34ef...
Oct 3, 05:37 AM	Registered	Detail text 5	127.0.0.1	ab12cd34ef...

AS Aarav Sharma

aarav@wallet.demo

Security page

Turn on two-factor sign-in (recommended)

The screenshot shows the IdentityWallet 'Security' page. On the left is a sidebar with navigation links: Dashboard, My credentials, Sharing & consent, Document vault, Security (highlighted), and Verify a credential. The main content area is titled 'Security' and includes a sub-header 'Sign-in protection, your keys and the tamper-evident activity log.' It features three main sections: 1. 'Two-factor sign-in' with a QR code and a 6-digit code input field. 2. 'Identity & keys' showing DID, Key fingerprint, Key type, and Last sign-in, along with a 'Change password' form. 3. 'Activity log' with a table of recent events.

Two-factor sign-in

1. Open Google Authenticator, Microsoft Authenticator or Authy and scan:

Can't scan? Enter this key: JBSW Y3DP EHPK 3PXP JBSW Y3DP EHPK 3PXP

2. Enter the 6-digit code it shows:

0 0 0 0 0 0

Confirm and turn on

Identity & keys

DID: did:ldw:df3d86746f9d0116315d16de12fec3a3eb628d6c

Key fingerprint: DF3D:8674:6F9D:0116

Key type: ECDSA P-256, private key encrypted at rest

Last sign-in: Oct 4, 05:37 PM

Change password

Current password:

New password: Repeat:

Update password

Activity log

Each entry stores the SHA-256 hash of the one before it, so edits to the log are detectable.

WHEN	EVENT	DETAIL	IP	HASH
Oct 4, 05:37 PM	Login	Detail text 0	127.0.0.1	ab12cd34ef...
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Oct 4, 03:13 AM	Share created	Detail text 2	127.0.0.1	ab12cd34ef...
Oct 3, 08:01 PM	Login failed	Detail text 3	127.0.0.1	ab12cd34ef...

Authenticator setup

- 1 Install **Google Authenticator**, **Microsoft Authenticator** or **Authy** on your phone.
- 2 Click **Set up authenticator app**.
- 3 Scan the QR code with the app (or type the key shown under it).
- 4 Type the 6-digit code from the app and click **Confirm and turn on**.

From now on, sign-in asks for the code after your password. To turn it off, enter your password and a current code, then click **Turn off**.

Change password: enter the current password and the new one twice, then **Update password**.

Activity log: the latest 100 events for your account. Each entry carries a hash of the previous one, so changes to the log can be detected.

5 Issuer Guide

The screenshot shows the IdentityWallet Issuer portal. On the left is a sidebar with 'IdentityWallet Digital Identity', 'Issue credentials', 'Security', and 'Verify a credential'. The main area is titled 'Issuer portal' and 'Issue signed credentials straight into people's wallets.' It features a 'Government issuer' card for 'National Identity Authority (Demo)' with an 'ACTIVE' status and a signing key. Below this is the 'Issue a credential' form with tabs for 'National Identity Card' and 'Student ID'. The form includes fields for 'Holder's wallet email' (person@example.com), 'Full name', 'Date of birth', and 'Validity' (3650 days). A 'Sign & issue' button is at the bottom. To the right, a 'What happens' section lists four steps. At the bottom, an 'Issued credentials' table shows three credentials: 'National Identity Card', 'Student ID', and 'Driving Licence', each with holder details, issue/expiration dates, and status (ACTIVE or REVOKED).

CREDENTIAL	HOLDER	ISSUED	EXPIRES	STATUS
National Identity Card cred_a1	Aarav Sharma aarav@wallet.demo	Sep 29, 2026	Aug 12, 2036	ACTIVE
Student ID cred_a2	Aarav Sharma aarav@wallet.demo	Sep 29, 2026	Nov 3, 2026	ACTIVE
Driving Licence cred_a3	Aarav Sharma aarav@wallet.demo	Sep 29, 2026	Aug 12, 2036	REVOKED Re-issued

Issuer portal

5.1 Issue a credential

- 1 Sign in as issuer staff (e.g. `registrar@identity.demo`). The top card shows your organisation and signing key.
- 2 Choose a credential type: National Identity Card, Driving Licence, Student ID, Employment Certificate, Health Insurance Card or Address Proof.
- 3 Enter the **holder's wallet email**. The person must already have a wallet account.
- 4 Fill the fields (* = required). Dates use the date picker.
- 5 Optional: click **Extra claim** to add custom details (name + value), and set **Validity** in days (0 = never expires; empty = template default).
- 6 Click **Sign & issue**. The credential appears instantly in the holder's wallet.

Automatic age claims

If you enter a date of birth, "Age over 18" and "Age over 21" are added automatically.

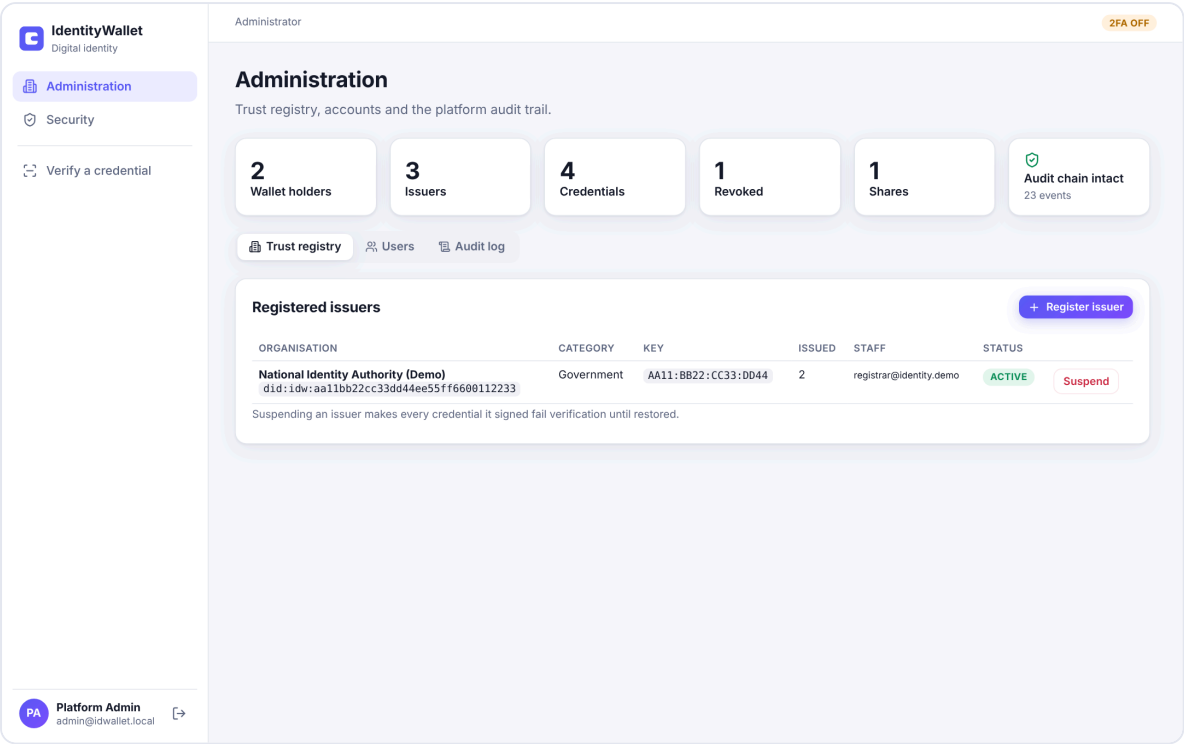
5.2 Revoke a credential

1. In **Issued credentials**, find the credential (use **Search**).
2. Click **Revoke**, enter a reason (e.g. "Issued in error") and confirm.
3. From now on every verifier sees it as revoked, and the holder can no longer share it.

Privacy note

Issuers see the list of what they issued and to whom, but not the claim values afterwards — those are encrypted for the holder.

6 Administrator Guide

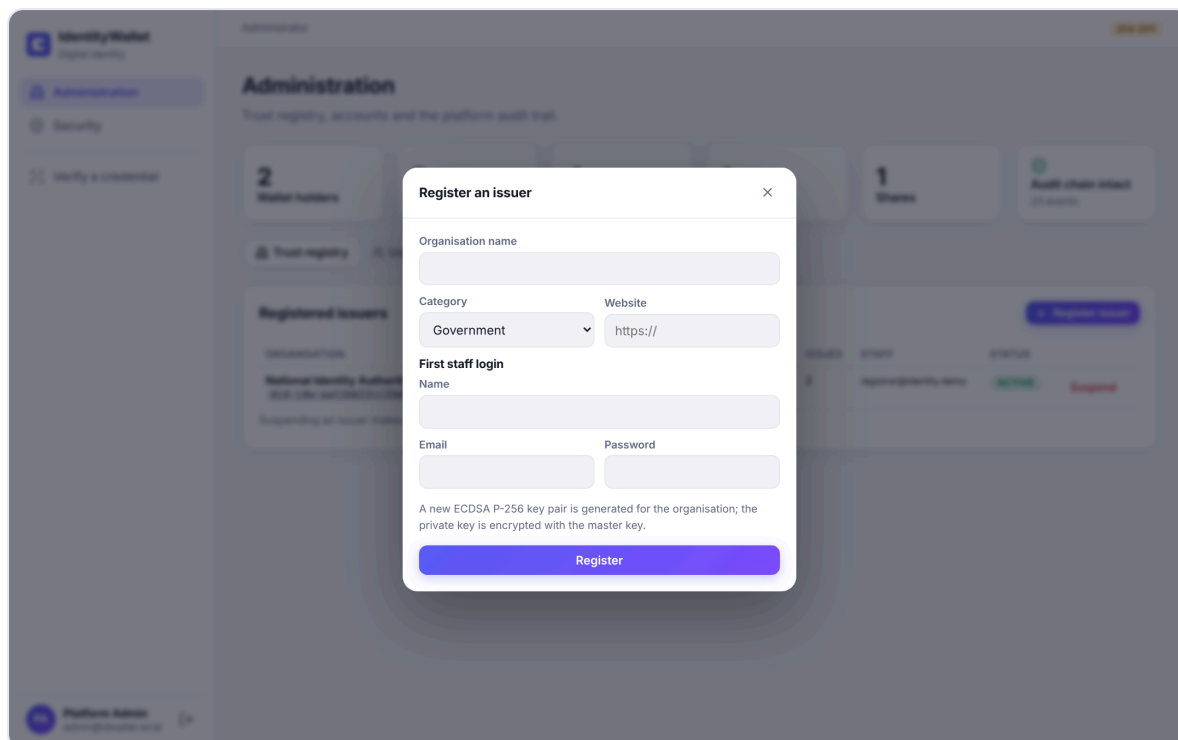


Administration

6.1 Statistics

The top row shows wallet holders, issuers, credentials, revoked credentials, shares and the **audit chain status** (intact / broken at event #N).

6.2 Register an issuer (trust registry)

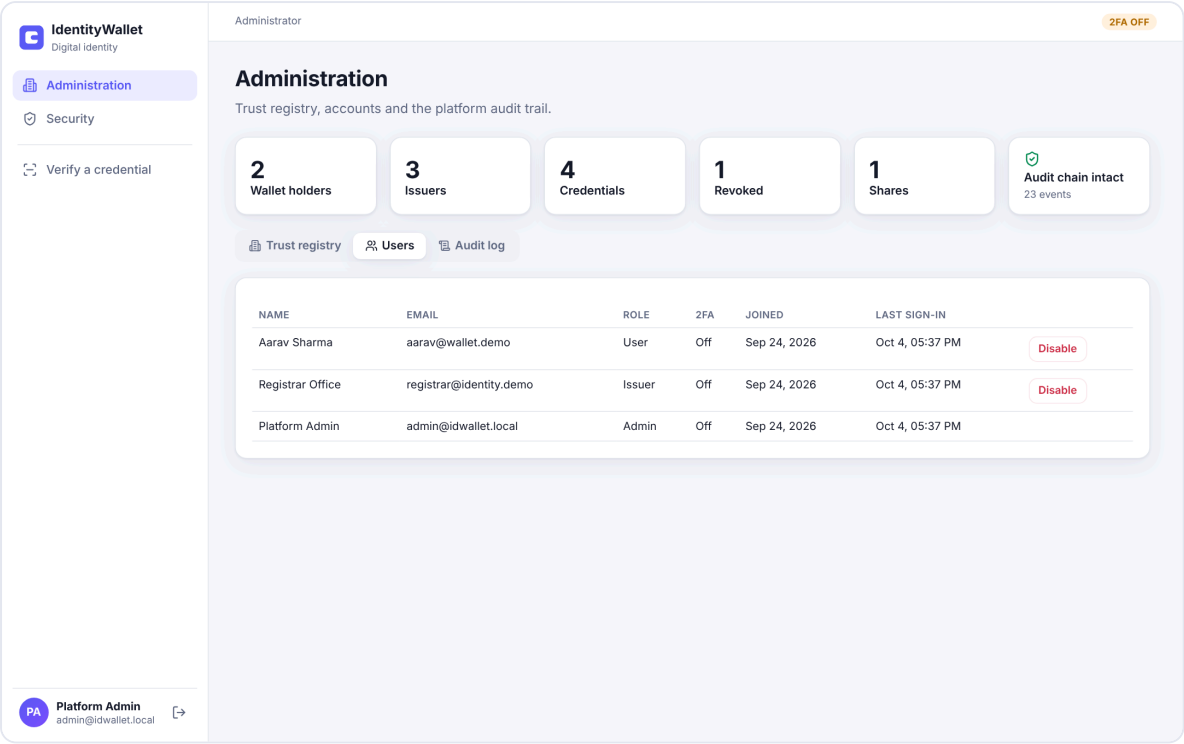
The image shows a screenshot of the IdentityWallet Administration interface. A modal window titled 'Register an issuer' is centered on the screen. The modal contains the following fields: 'Organisation name' (text input), 'Category' (dropdown menu with 'Government' selected), 'Website' (text input with 'https://' pre-filled), 'First staff login' section with 'Name', 'Email', and 'Password' (all text inputs). Below these fields, a small text note states: 'A new ECDSA P-256 key pair is generated for the organisation; the private key is encrypted with the master key.' At the bottom of the modal is a blue 'Register' button. The background of the interface is dimmed, showing the 'Administration' section with options like 'Trust registry', 'Accounts', and 'Platform audit trail'.

Register an issuer

1. On **Trust registry**, click **Register issuer**.
2. Enter organisation name, category and website.
3. Enter the first staff member's name, email and password.
4. Click **Register**. A new ECDSA signing key pair is generated for the organisation.

Suspend / Restore: suspending an issuer makes every credential it signed fail verification and stops new issuing. Restore reverses this.

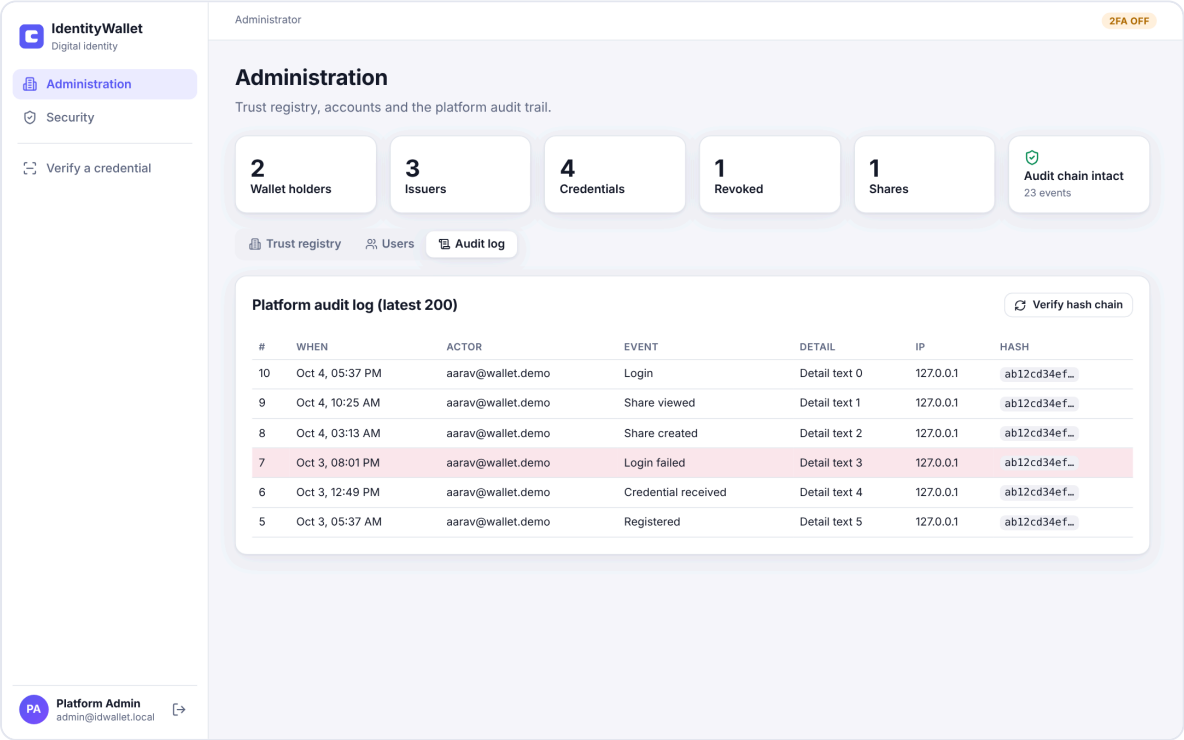
6.3 Users



Users

Lists every account with role, 2FA status, join date and last sign-in. **Disable** blocks a user from signing in; **Enable** restores access. Admins cannot disable themselves.

6.4 Audit log

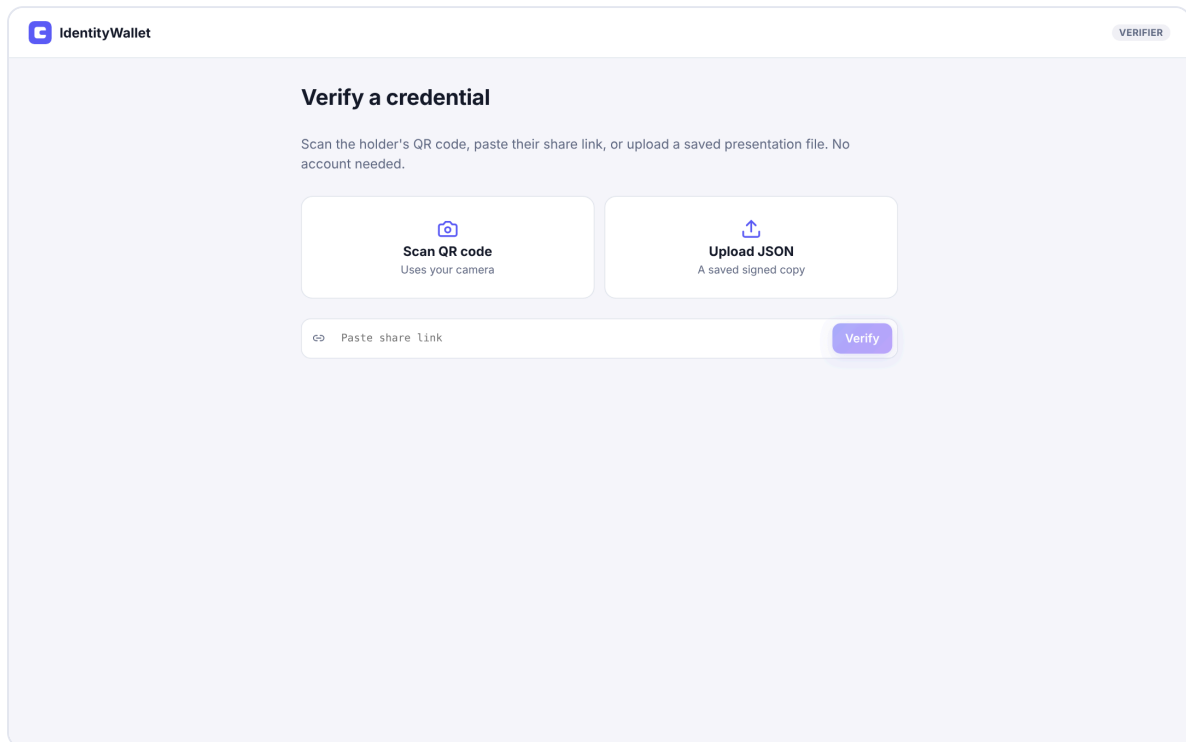


Audit log

Shows the latest 200 platform events. Failed logins are highlighted. **Verify hash chain** re-checks every event; if anyone edited or deleted a row in the database, the status shows the event number where the chain broke.

7 Verifier Guide (no account needed)

7.1 Ways to verify



IdentityWallet VERIFIER

Verify a credential

Scan the holder's QR code, paste their share link, or upload a saved presentation file. No account needed.

**Scan QR code**
Uses your camera

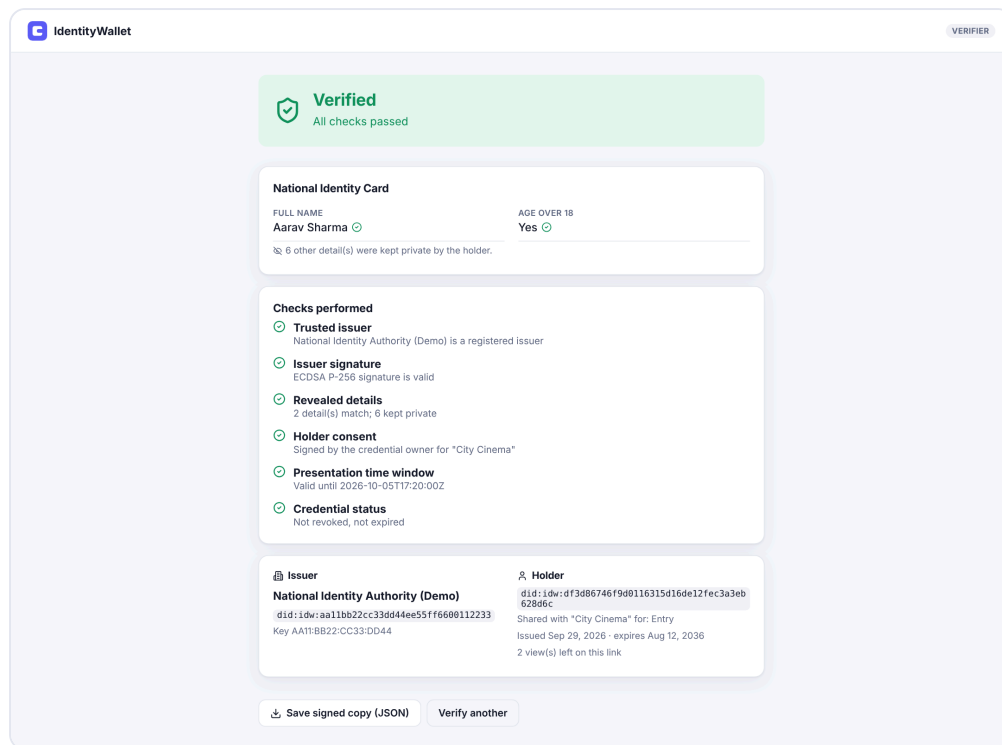
**Upload JSON**
A saved signed copy

Verify

Verifier page

Method	How
Open the link	Click the link the holder sent. The result appears directly
Scan QR code	Go to <code>/verify</code> , click Scan QR code , allow the camera and point it at the holder's QR
Paste link	Paste the link into the box and click Verify
Upload JSON	Upload a previously saved signed copy

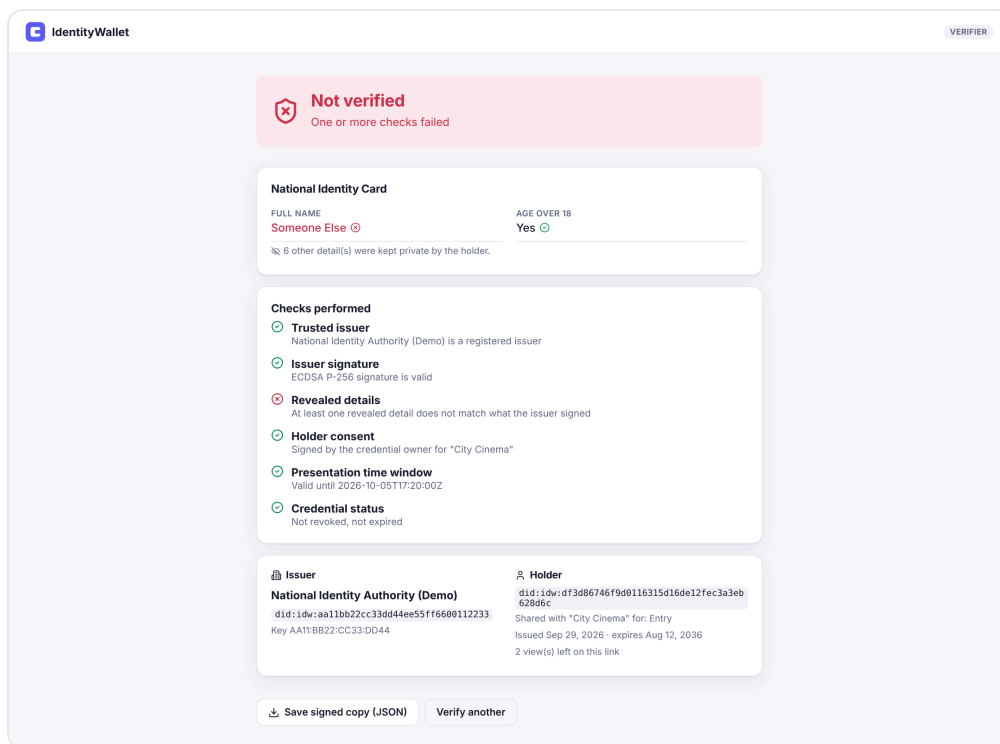
7.2 Reading the result



Verified result

- **Verified (green)** — all checks passed. Only the shared details are listed; the line below tells how many were kept private.
- **Not verified (red)** — at least one check failed. Read the red item to see why.

Check	Passes when
Trusted issuer	The issuer is registered and not suspended
Issuer signature	The credential has not been modified since signing
Revealed details	Every shown detail matches what the issuer signed
Holder consent	The owner signed this share for this recipient
Presentation time window	The share has not expired
Credential status	Not revoked and not expired



A tampered presentation is rejected

Save signed copy (JSON) downloads the presentation so you can re-check it later with **Upload JSON** — for example to confirm it was not revoked afterwards.

8 Troubleshooting and FAQ

8.1 Common problems

Problem	Solution
<code>mvn</code> is not recognized	Use <code>.\mvnw.cmd</code> instead of <code>mvn</code> (it downloads Maven for you)
<code>java</code> is not recognized	Install JDK 21 (adoptium.net), tick "Set JAVA_HOME", open a new PowerShell
<code>npm</code> is not recognized	Install Node.js LTS from nodejs.org, open a new window
Website shows "Request failed (500)" or "Something went wrong" on sign-in	The backend is not running — start it in Window 1 and wait for "Started IdWalletApplication"
Port 8080 was already in use	Another program uses the port. Close the old backend window, or restart the PC
Account locked	Wait 5 minutes after 5 wrong passwords
2FA code always wrong	Your phone's time must be correct: turn on automatic date & time
Lost authenticator phone	Delete the <code>backend\data</code> folder to reset the demo (all data is lost)
Share link says "not valid"	Check the full link was copied; it may have been stopped or the wallet reset
"This link has reached its view limit"	Create a new share with a higher limit
QR scanning doesn't start	Allow camera access in the browser, or paste the link instead
File upload refused	Only real PDF/PNG/JPEG/WEBP files up to 5 MB are allowed
Everyone signed out after restarting backend	Normal: the session key is random per start. Set <code>idwallet.jwt-secret</code> to keep sessions

8.2 FAQ

Can a holder fake a credential? No. Changing any value breaks the issuer's signature or the detail's hash, and verification fails.

Can a verifier see my hidden details? No. Hidden details are never sent; only salted hashes are in the signed data, and they cannot be reversed.

Where is my data stored? In the database in encrypted form. The decryption master key is stored separately in `backend\data\master.key` (or an environment variable).

Can I open share links on my phone? Yes, if the phone can reach your computer. For a demo, use a tunnel (e.g. `cloudflared tunnel --url http://localhost:5173`) and set `idwallet.public-url` to the tunnel address.

How do I build a single runnable file? In `frontend` run `npm run build:jar` , then in `backend` run `.\mvnw.cmd package` and start `java -jar target\identity-wallet-1.0.0.jar` . Everything then runs on `http://localhost:8080` (set `idwallet.public-url=http://localhost:8080`).